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(12) **United States Patent**
Komarasamy et al.

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(45) **Date of Patent:** ***Aug. 26, 2025**

(54) **SHEAR-ASSISTED EXTRUSION
ASSEMBLIES AND METHODS**

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patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

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(65) **Prior Publication Data**

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Related U.S. Application Data

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Sep. 14, 2022, now Pat. No. 11,919,061.

(60) Provisional application No. 63/244,632, filed on Sep.
15, 2021.

(51) **Int. Cl.**
B21C 23/04 (2006.01)

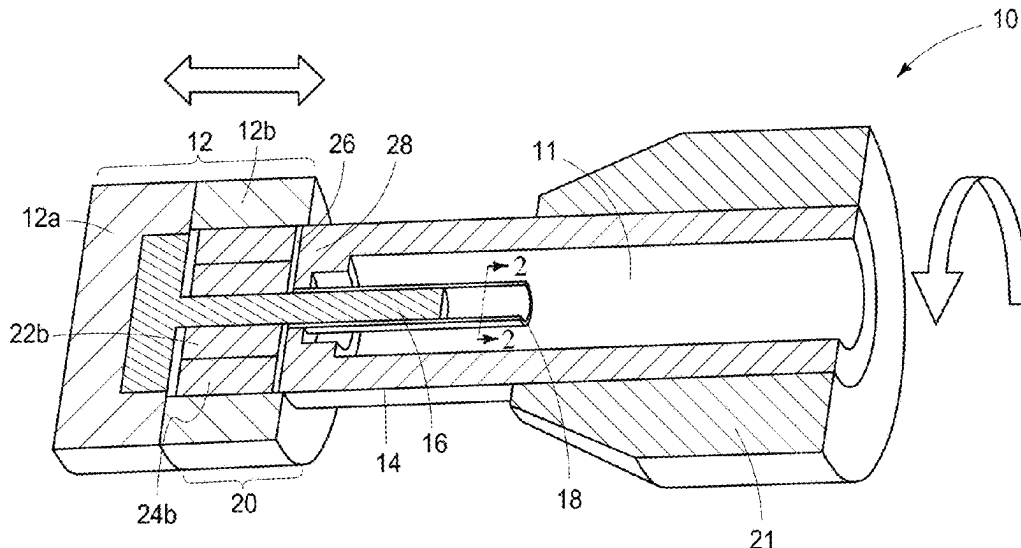
(52) **U.S. Cl.**
CPC **B21C 23/04** (2013.01)

(58) **Field of Classification Search**
CPC B21C 23/02; B21C 23/04; B21C 23/022
See application file for complete search history.

(57) **ABSTRACT**

Shear-assisted extrusion assemblies are provided. The assemblies can include a billet containing assembly containing a billet comprising a billet outer material and a billet inner material in at least one cross-section; a tool operably engaged with the billet; an extrudate receiving channel configured to receive extrudate from the tool, wherein the extrudate comprises extruded outer material and extruded inner material in at least one cross-section, the extruded outer material being the same material as the billet outer material, and the extruded inner material being the same as the billet inner material. Methods for producing multi-material shear-assisted extrudate are also provided.

20 Claims, 23 Drawing Sheets



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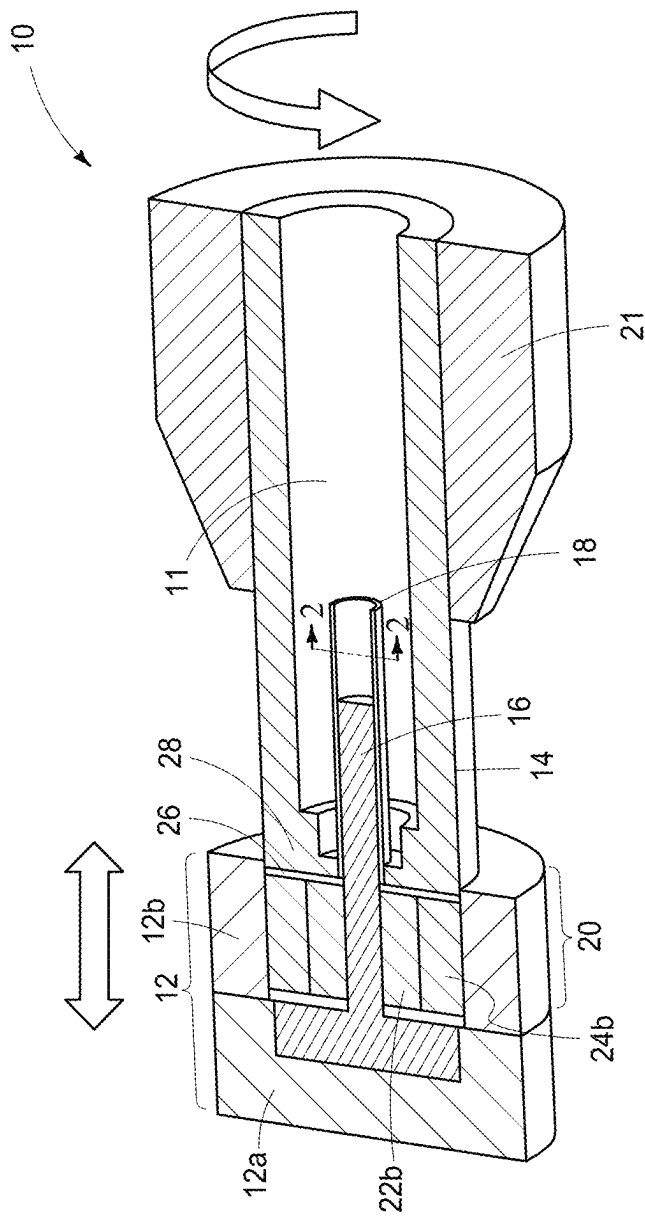


FIG. 1

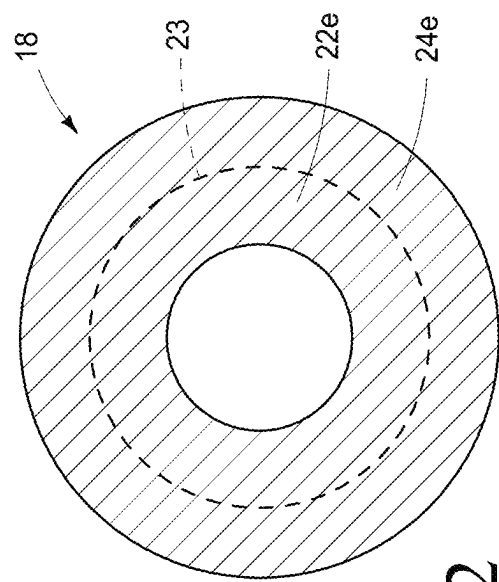


FIG. 2

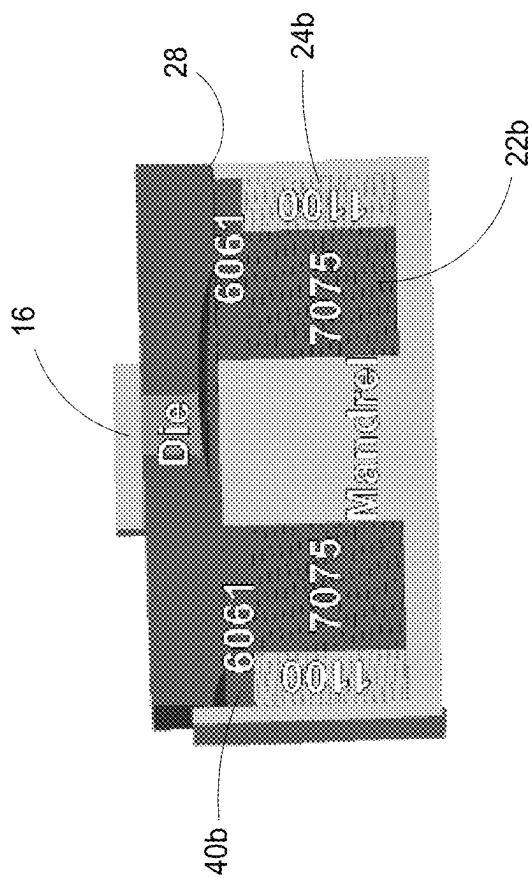


FIG. 3

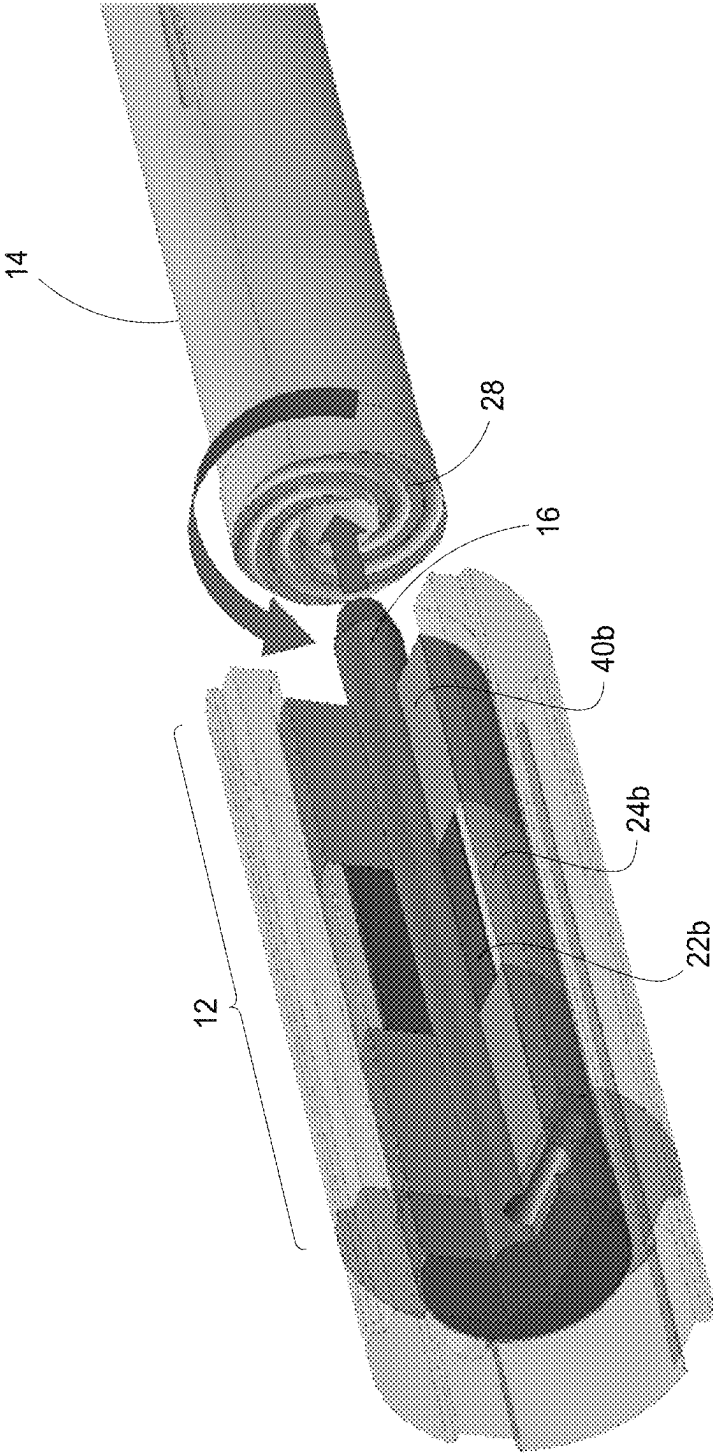


FIG. 4

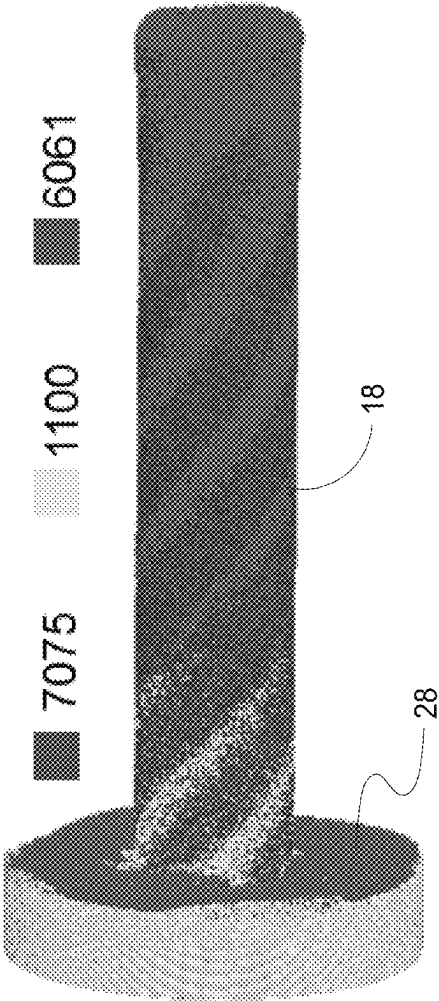
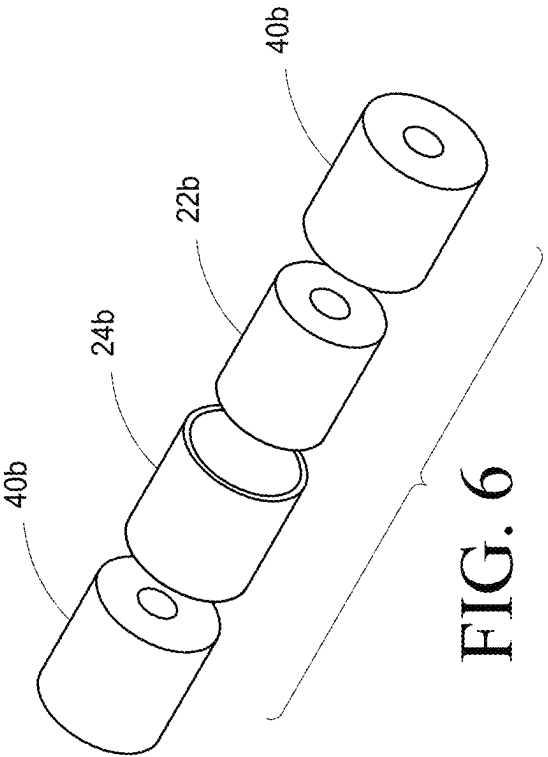
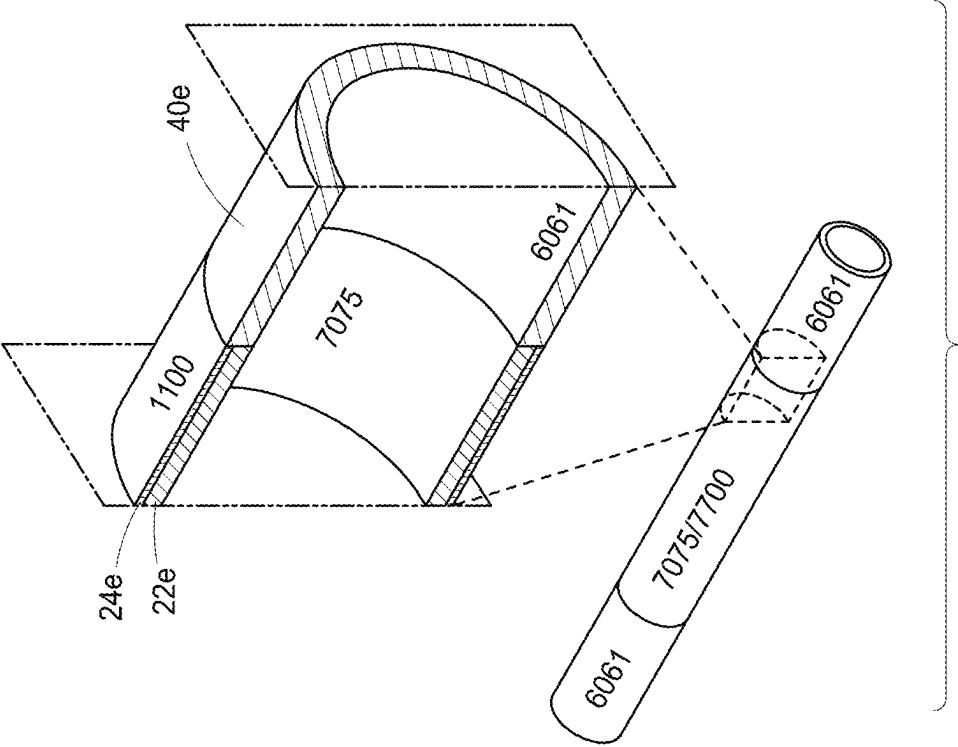


FIG. 5



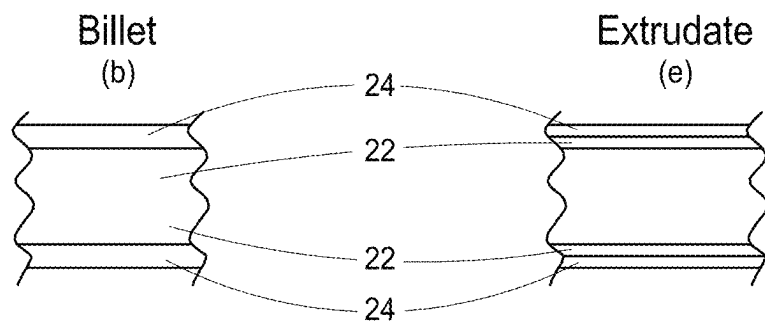


FIG. 8

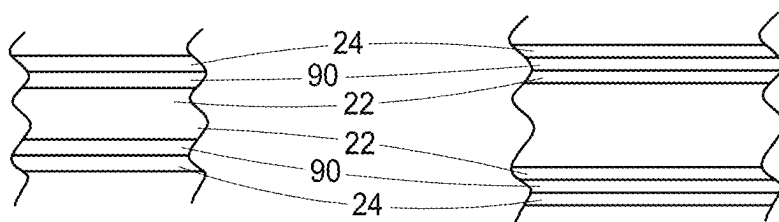


FIG. 9

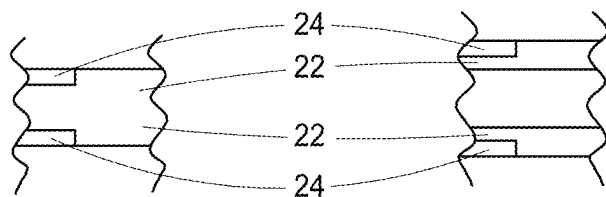


FIG. 10

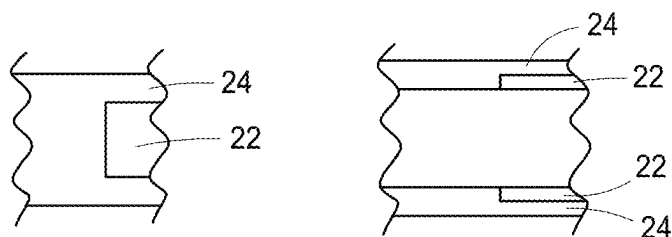


FIG. 11

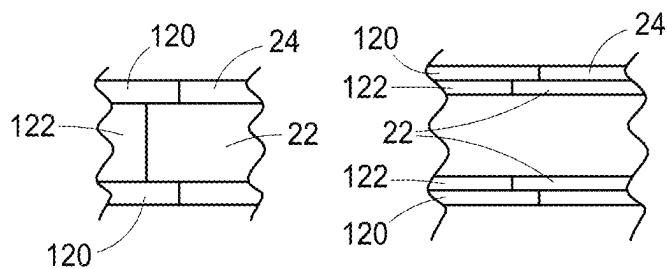


FIG. 12

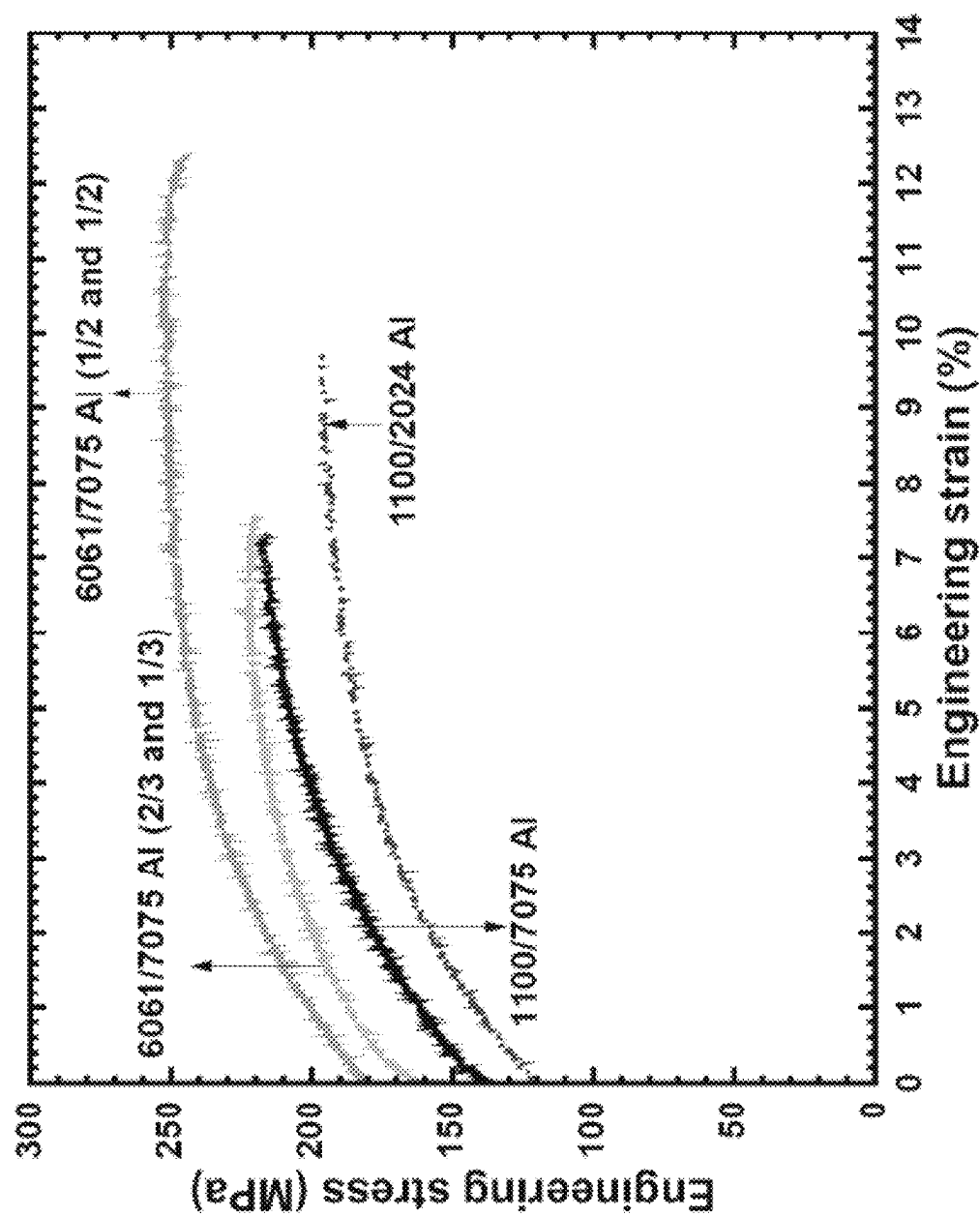


FIG. 13

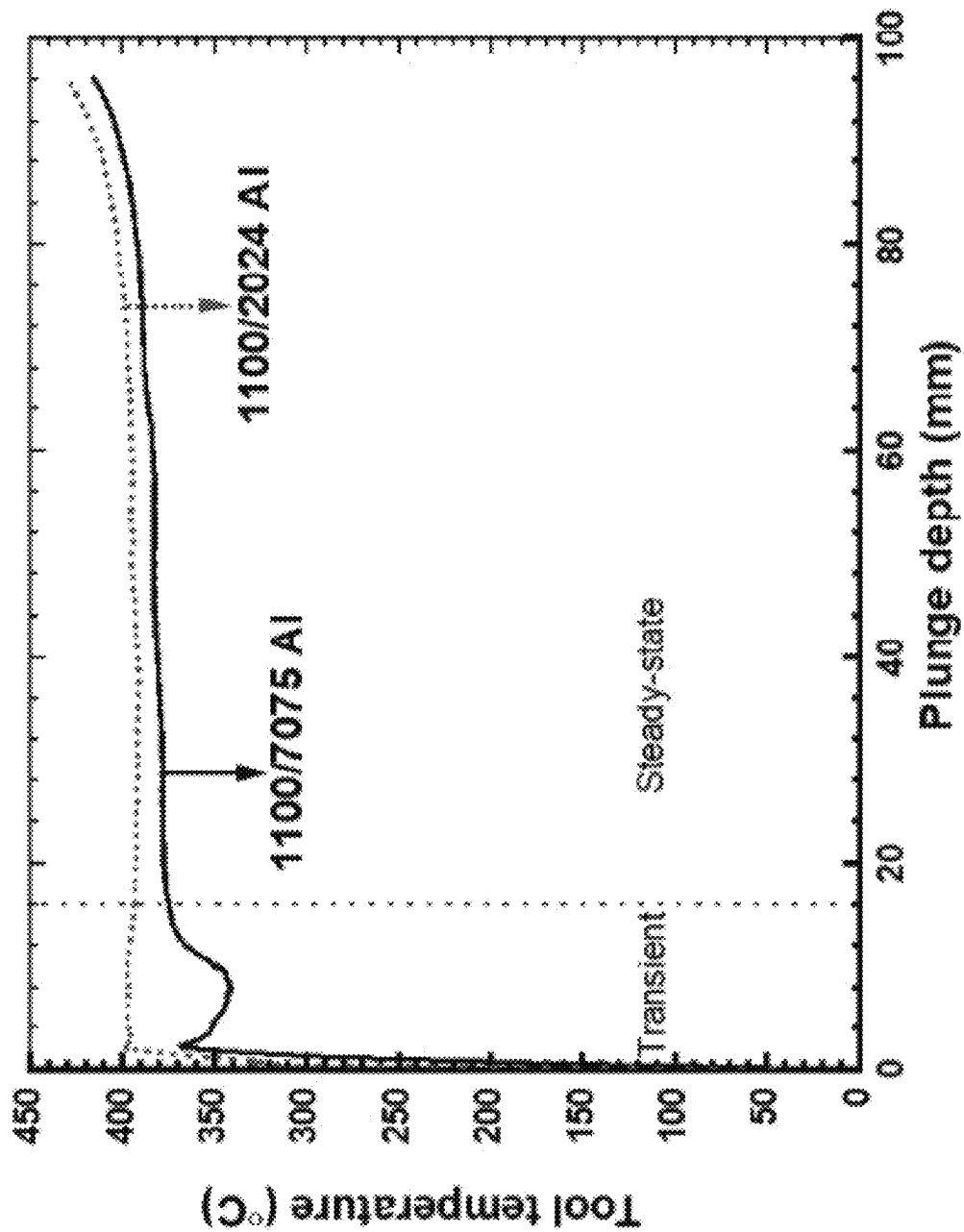


FIG. 14

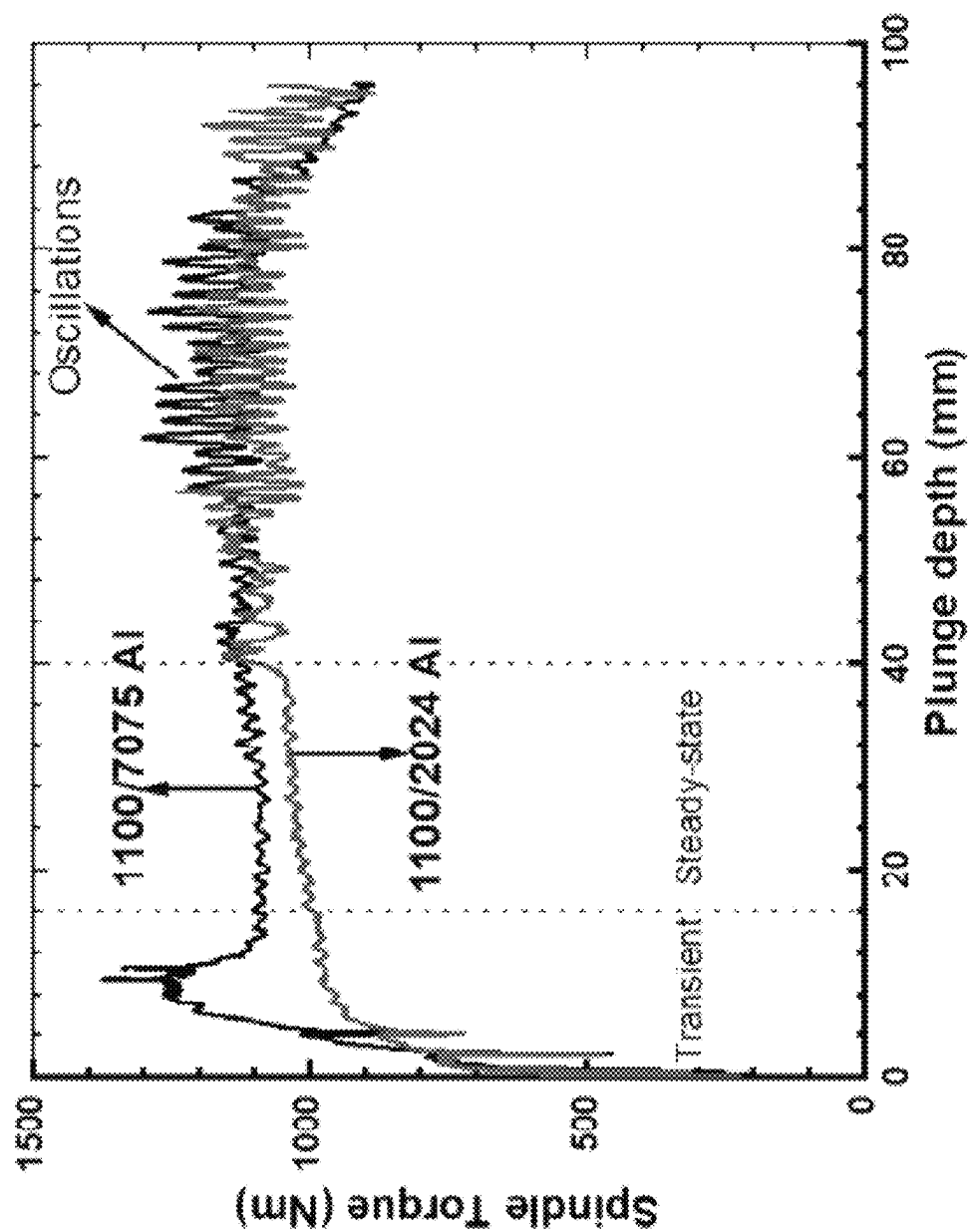


FIG. 15

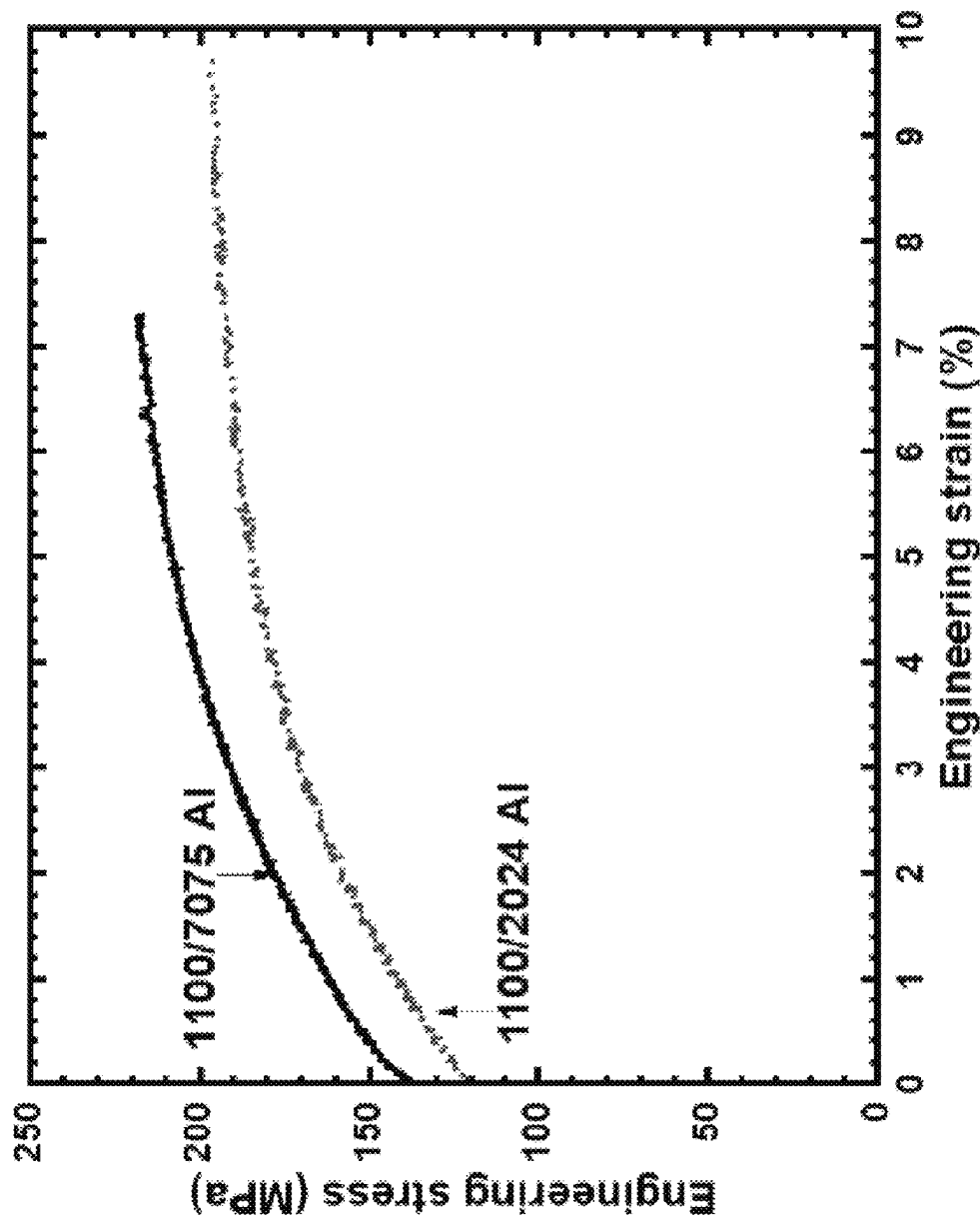


FIG. 16

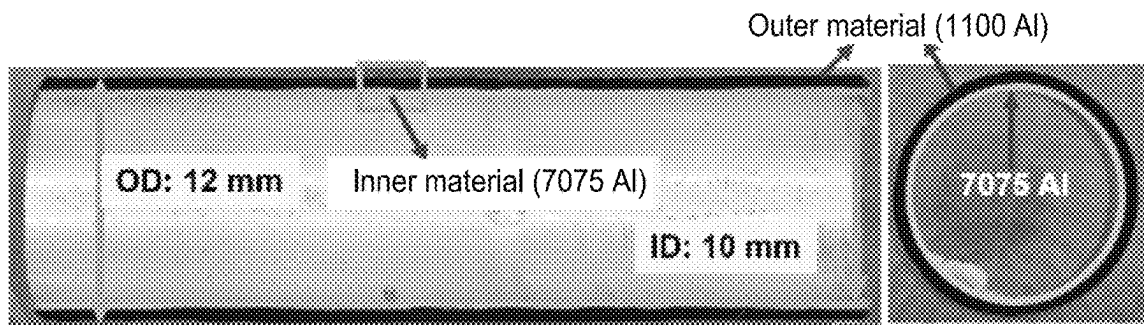


FIG. 17A

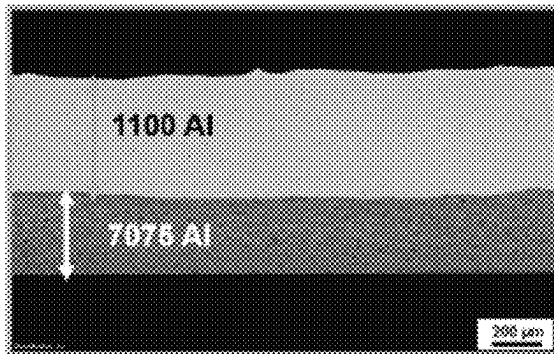


FIG. 17B

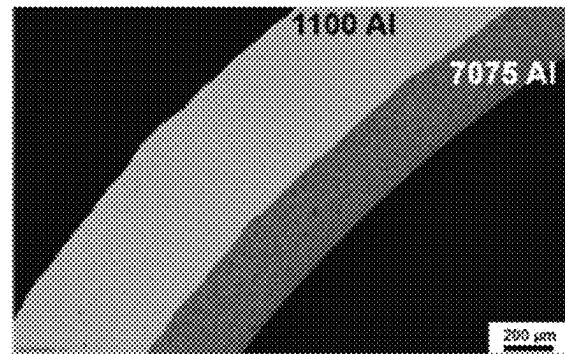


FIG. 17C

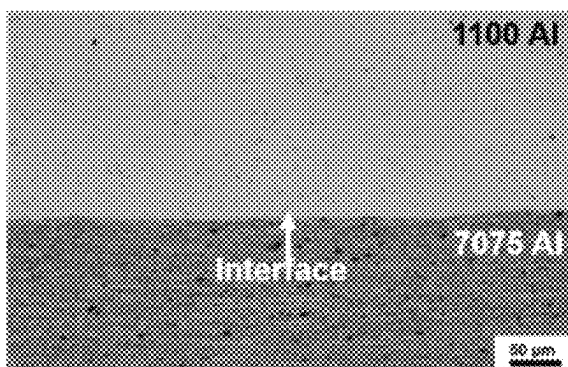


FIG. 17D

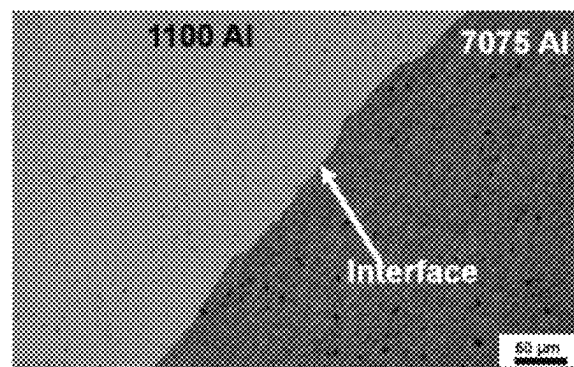


FIG. 17E

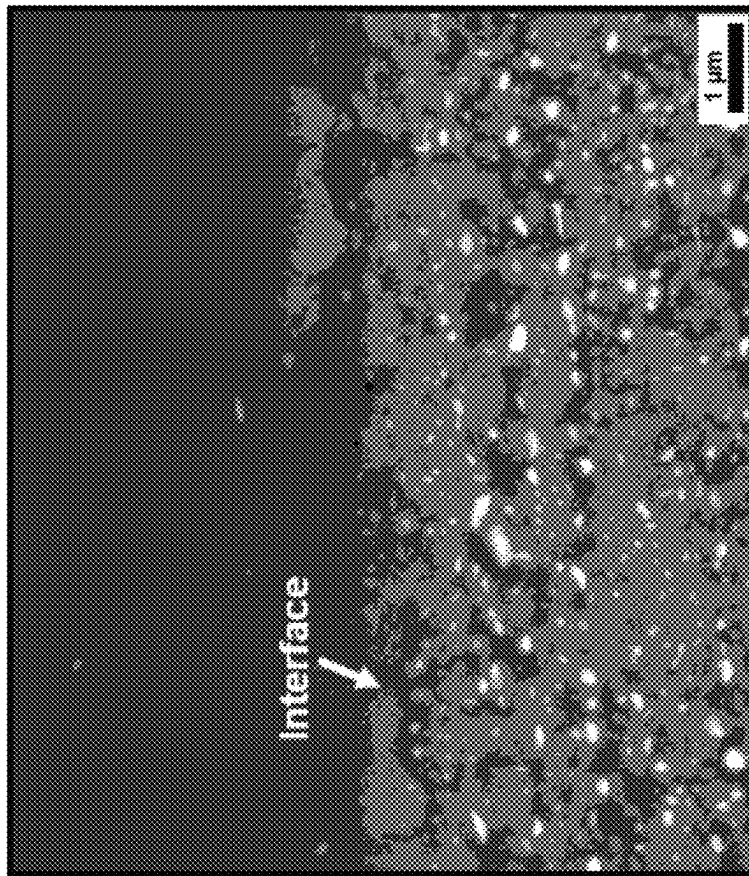


FIG. 18B

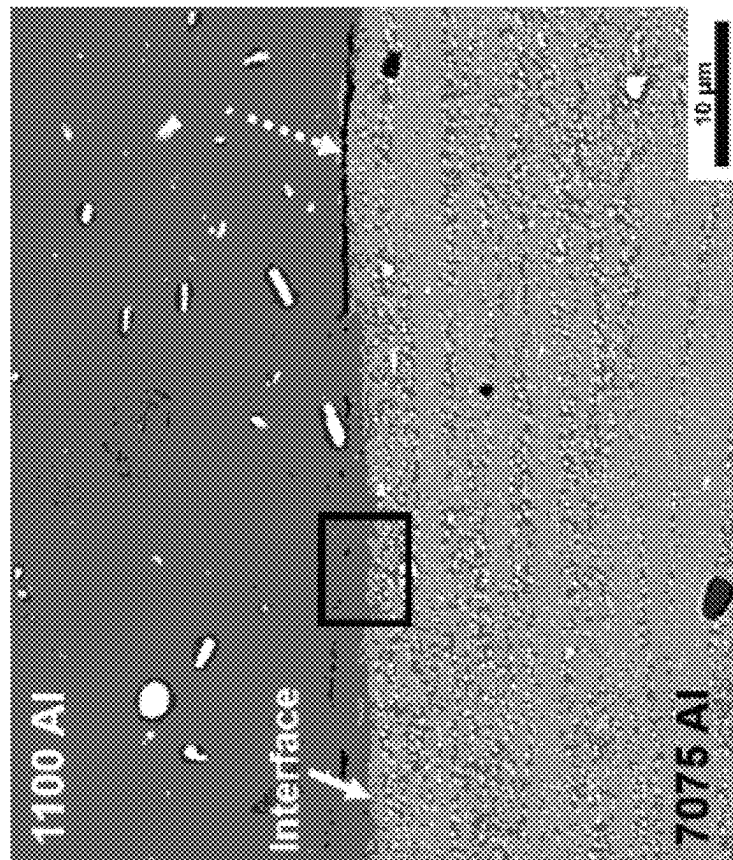


FIG. 18A

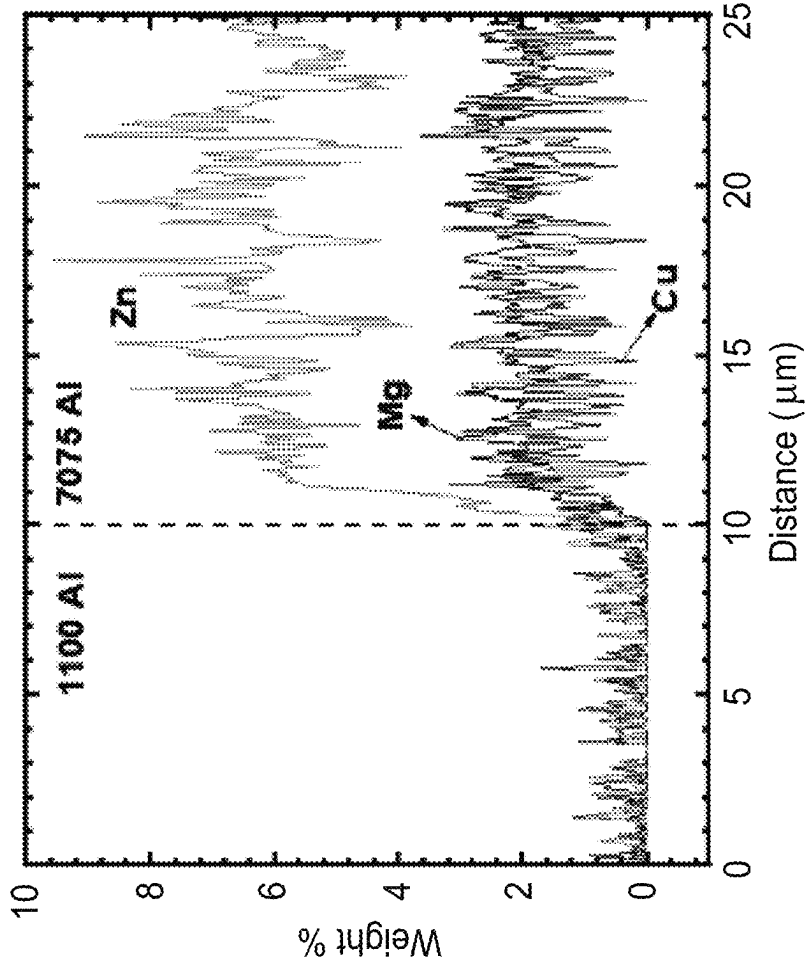


FIG. 18C

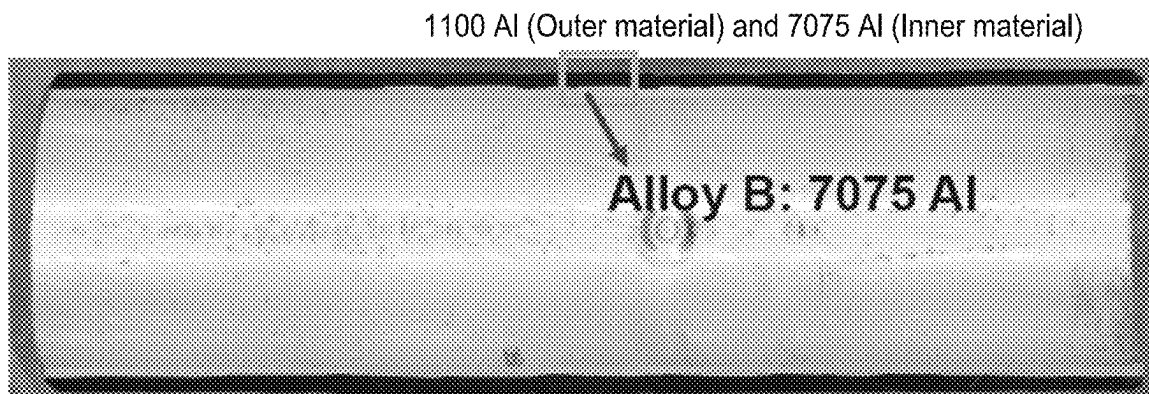


FIG. 19A

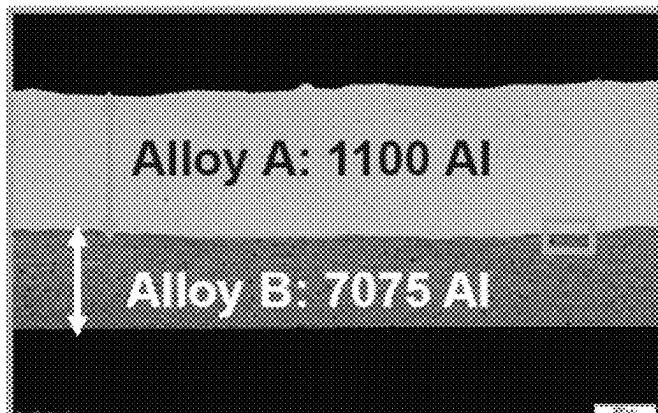


FIG. 19B

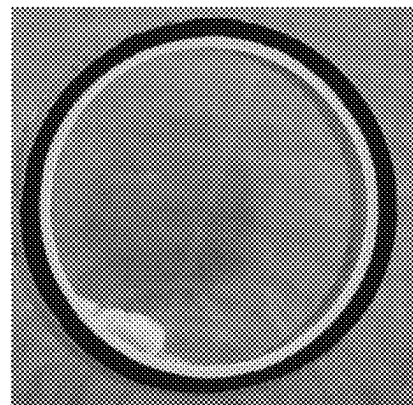


FIG. 19C

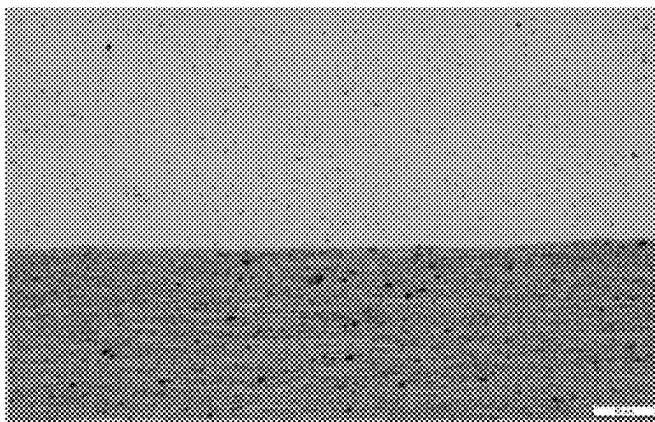


FIG. 19D

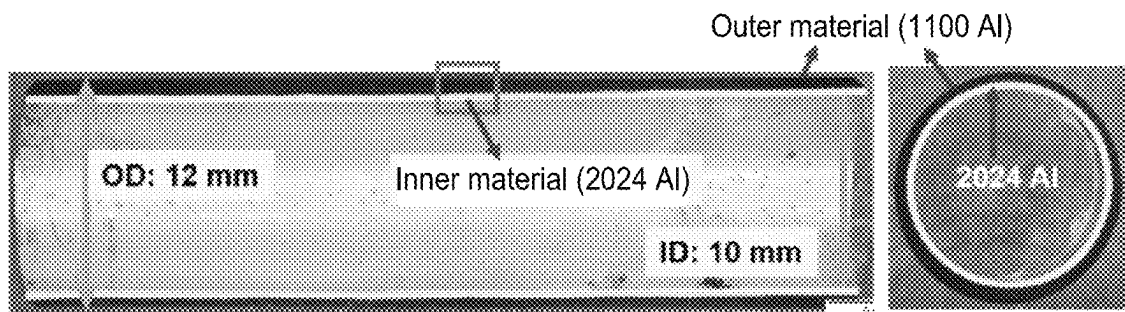


FIG. 20A

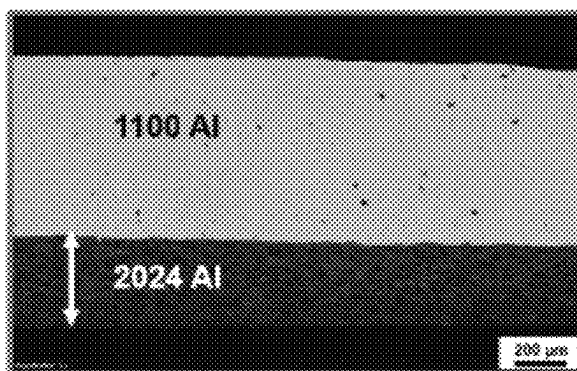


FIG. 20B

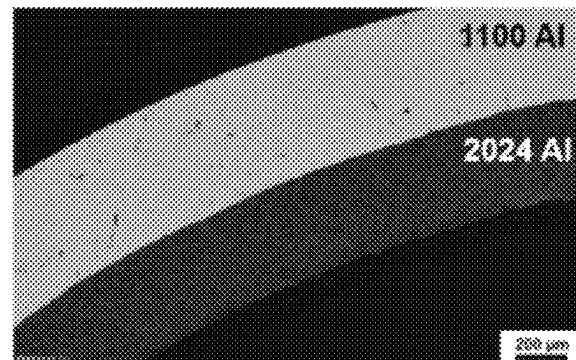


FIG. 20C

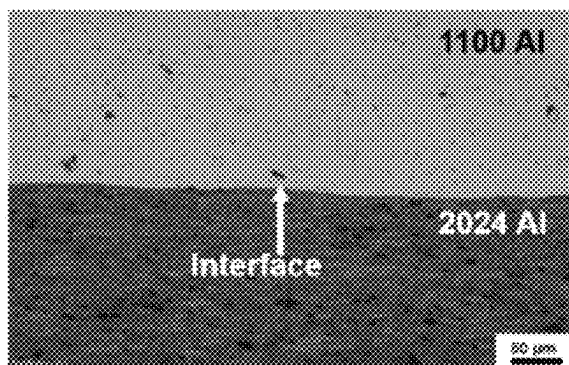


FIG. 20D

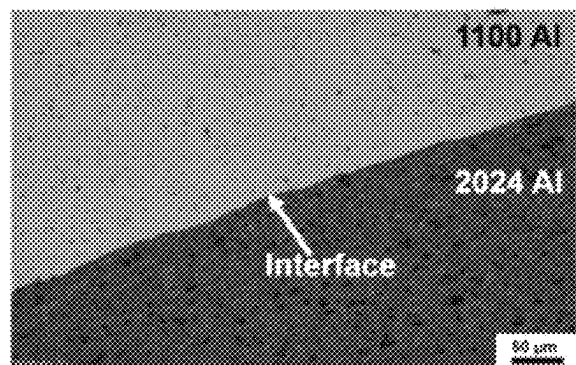


FIG. 20E

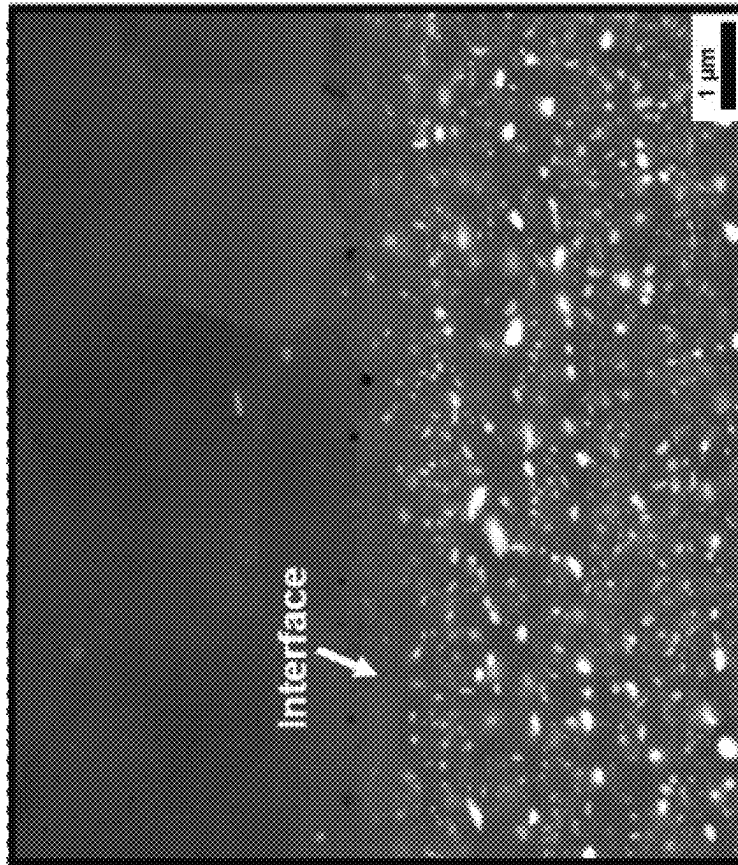


FIG. 21B

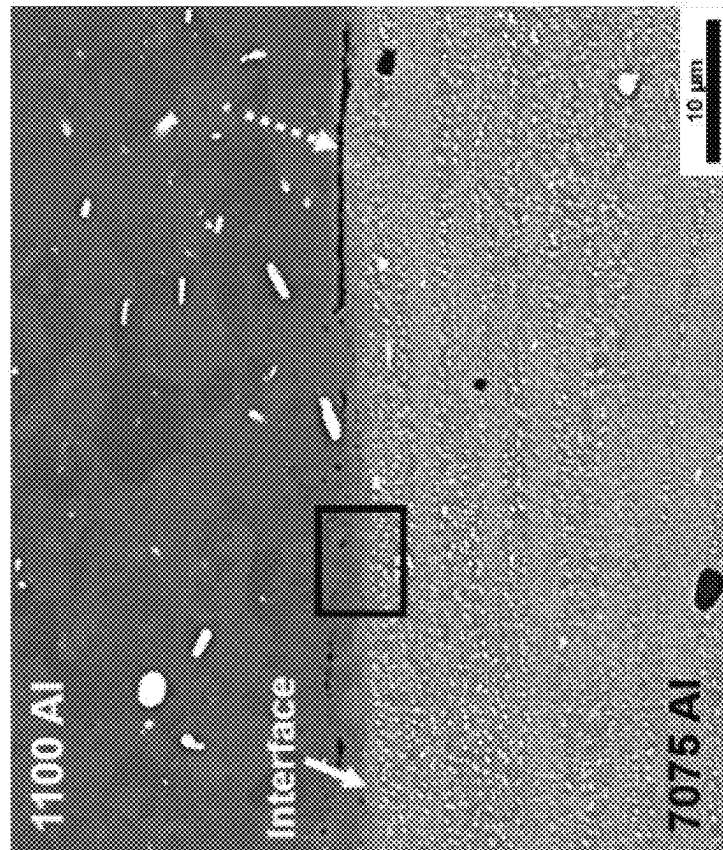


FIG. 21A

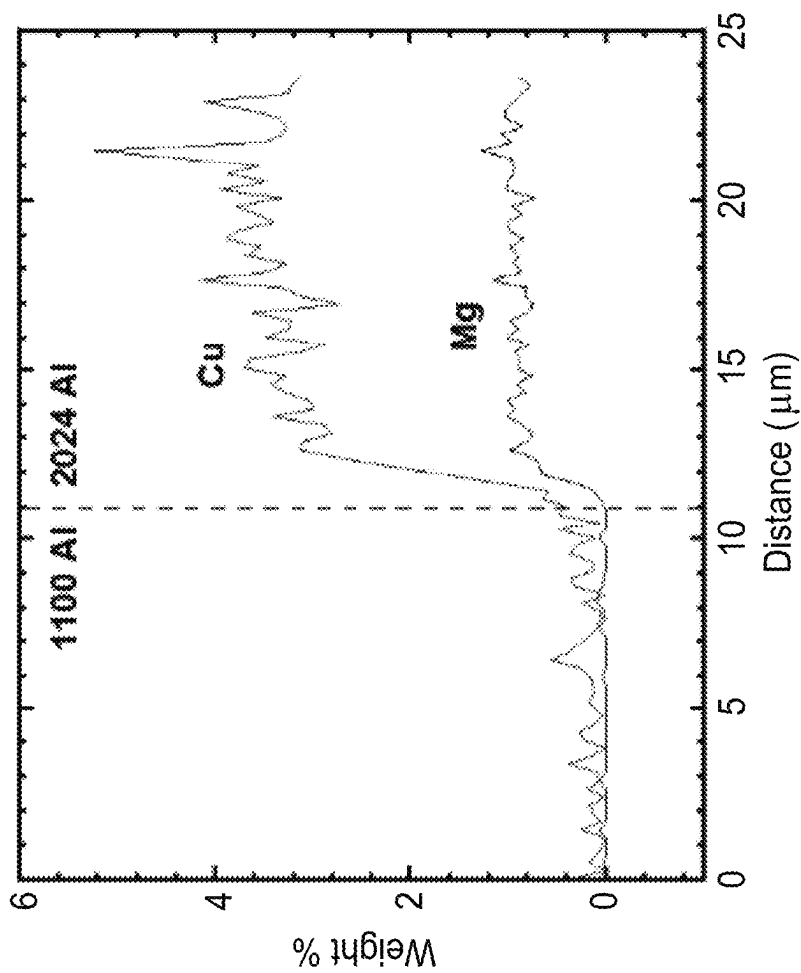


FIG. 21C

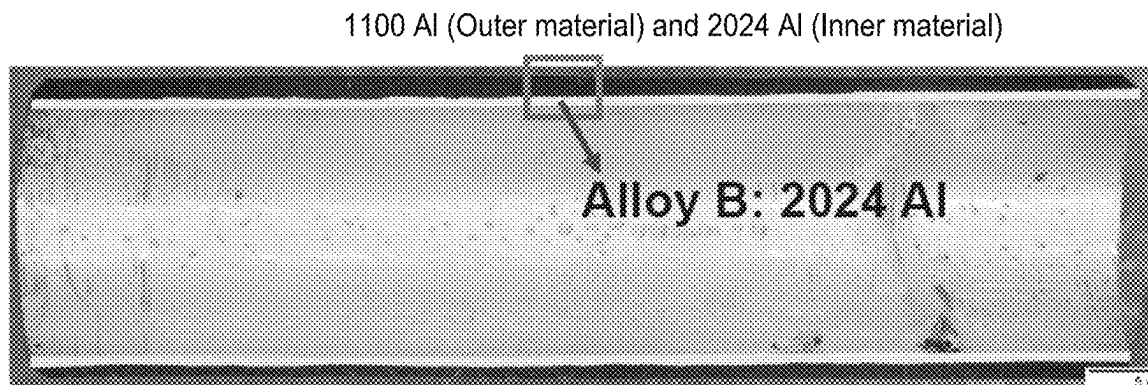


FIG. 22A

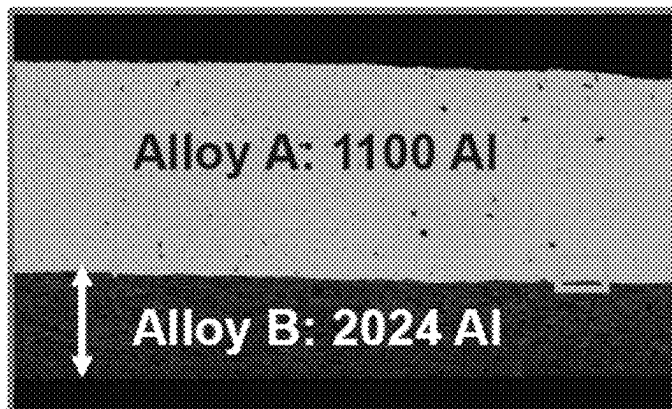


FIG. 22B

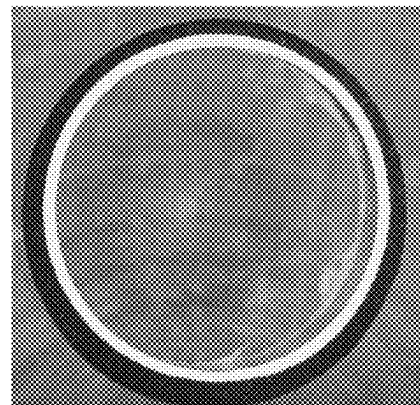


FIG. 22C

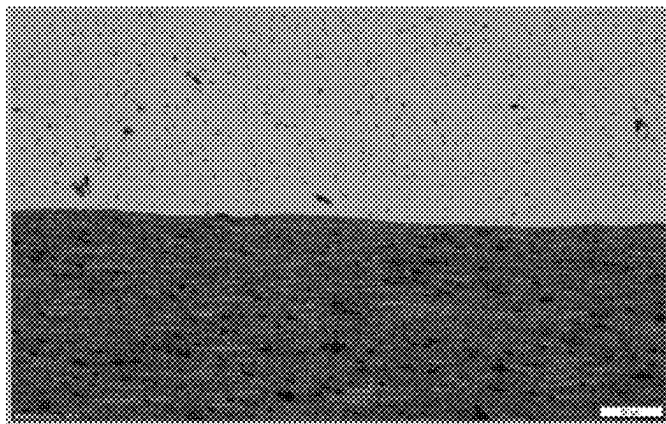


FIG. 22D

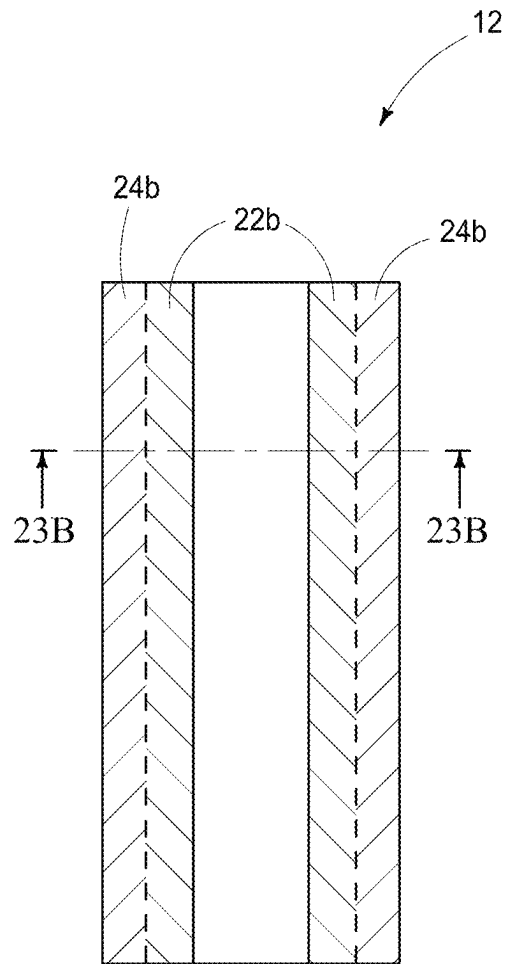


FIG. 23A

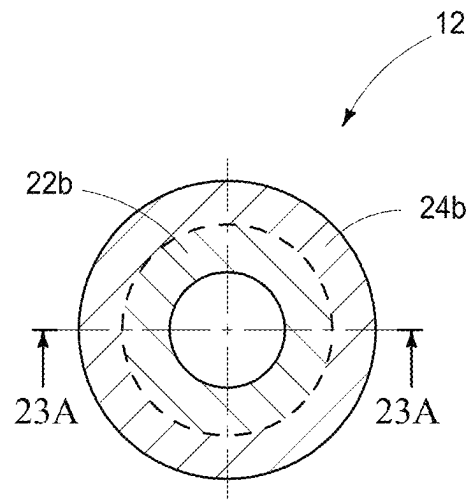


FIG. 23B

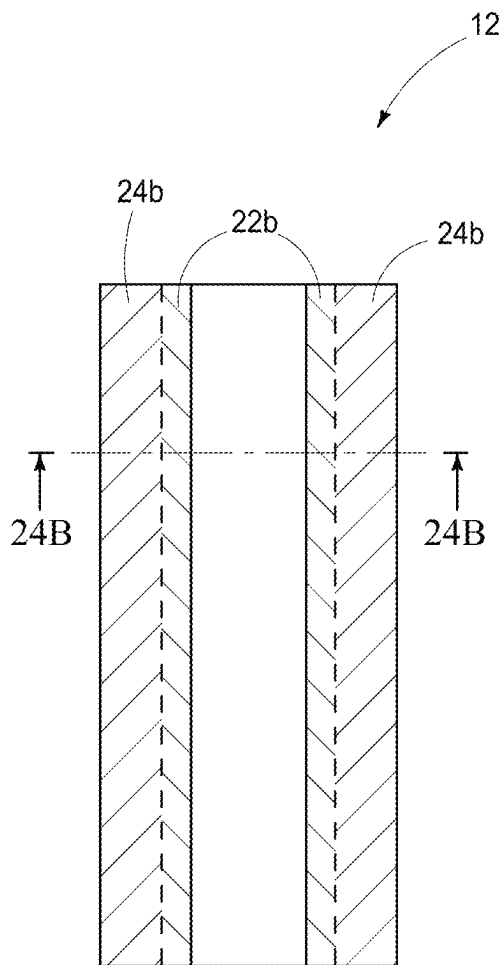


FIG. 24A

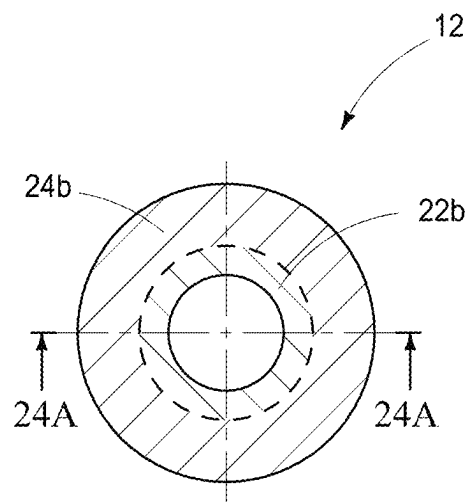


FIG. 24B

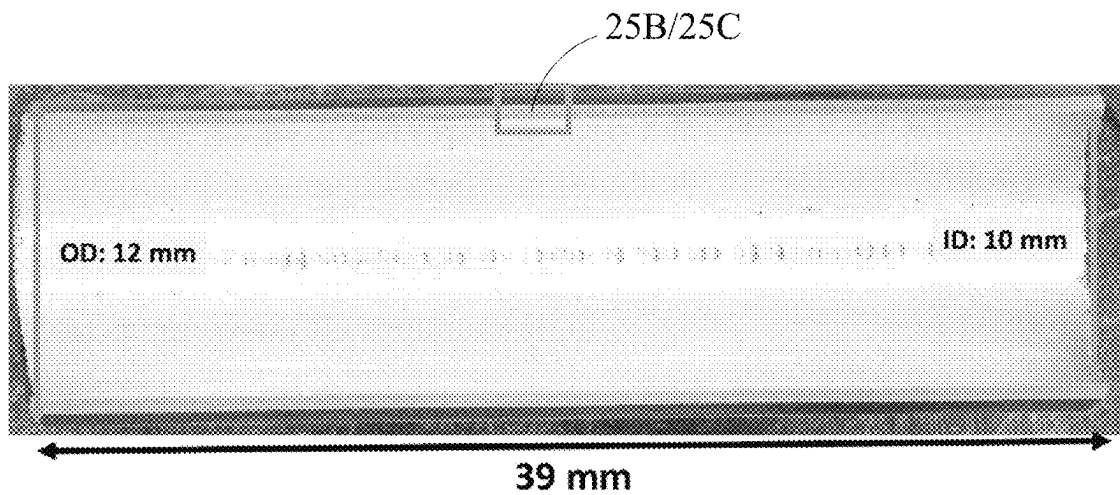


FIG. 25A

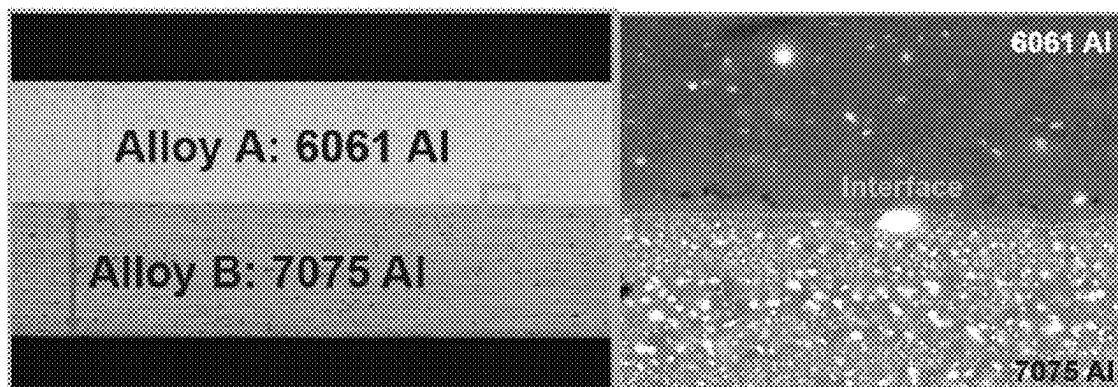


FIG. 25B

FIG. 25C

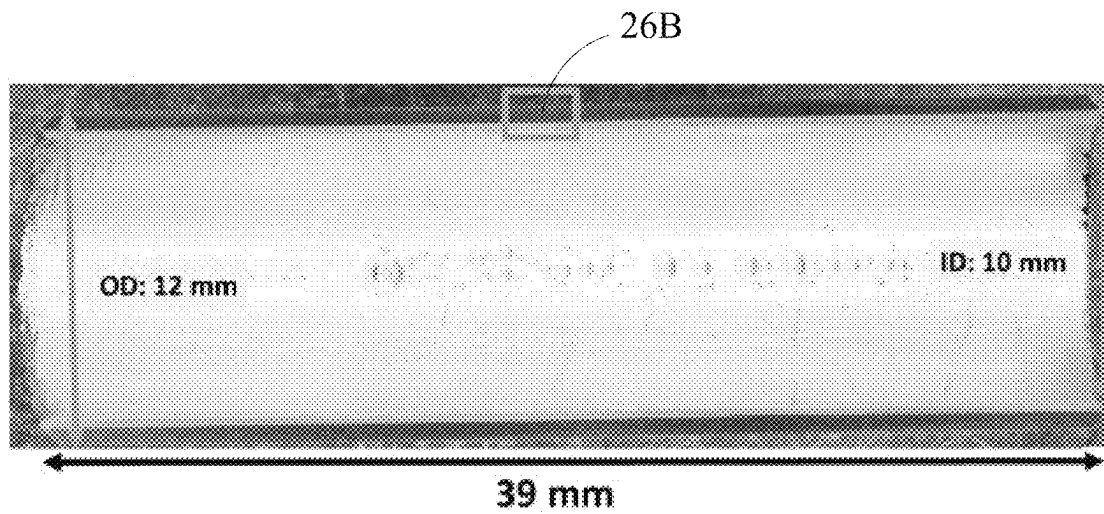


FIG. 26A

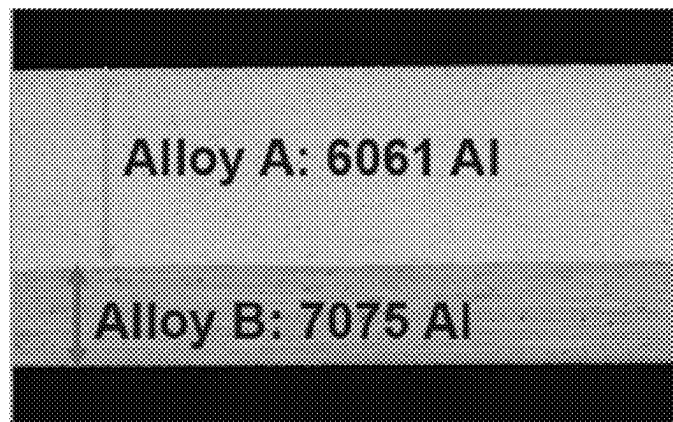


FIG. 26B

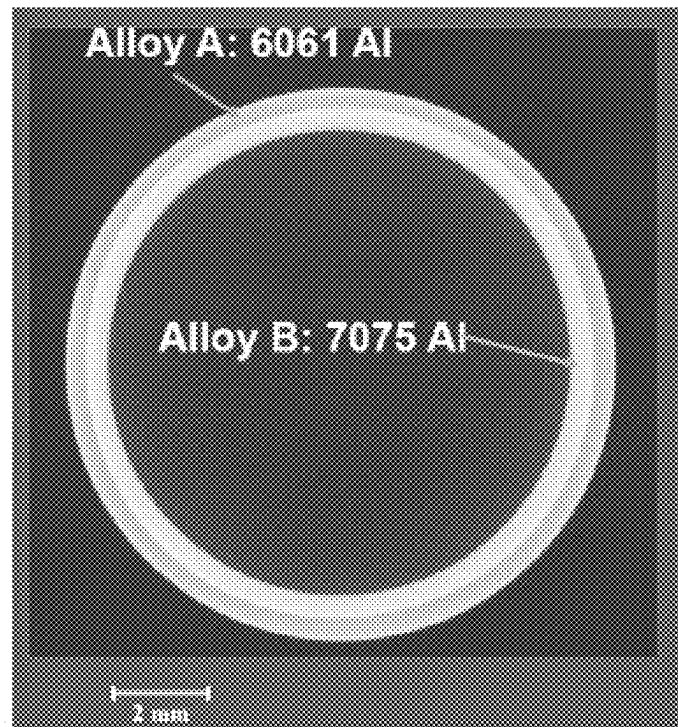


FIG. 27

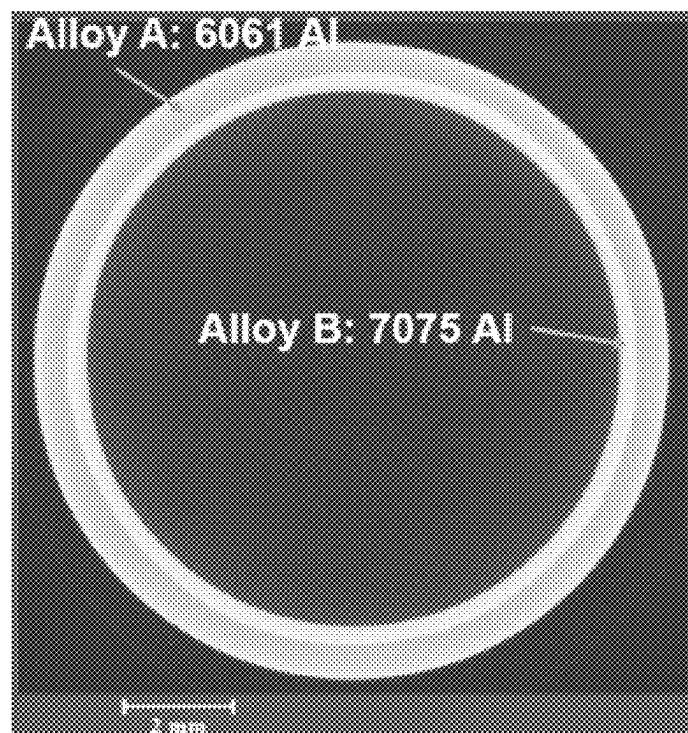


FIG. 28

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**SHEAR-ASSISTED EXTRUSION
ASSEMBLIES AND METHODS****CROSS REFERENCE TO RELATED
APPLICATION**

This application is a continuation of U.S. patent application Ser. No. 17/944,932 filed Sep. 14, 2022, which claims priority to and the benefit of U.S. Provisional Patent Application Ser. No. 63/244,632 filed Sep. 15, 2021, entitled “Co-Extrusion/Cladding of Dissimilar AL Alloys Via Shear Assisted Processing and Extrusion”, the entirety of each of which is incorporated by reference herein.

**STATEMENT AS TO RIGHTS TO INVENTIONS
MADE UNDER FEDERALLY-SPONSORED
RESEARCH AND DEVELOPMENT**

This invention was made with Government support under Contract DE-AC05-76RL01830 awarded by the U.S. Department of Energy. The Government has certain rights in the invention.

TECHNICAL FIELD

The present disclosure relates to extrusion assemblies and methods. Particular embodiments include shear-assisted extrusion assemblies and methods that can be used to create multi-metallic materials.

BACKGROUND

Co-extrusion via hydrostatic extrusion and indirect extrusion is either expensive or results in non-uniform material flow. Furthermore, elaborate billet fabrication steps, requirement of specific area ratio between core and the sleeve, difficulty in co-extruding 1100 and 7075Al, and billet pre-heating are typically employed to obtain desirable bonding at the interface. Thermal spray coatings are limited to thin coatings which might not be beneficial for long-term applications due to cracking and spallation, and explosive bonding has safety and material constraints.

A few research level solutions that can never be applied at a manufacturing scale are fabrication of bi-metallic laminates using equal channel angular pressing and/or high pressure torsion.

Therefore, for increased energy efficiency and reduced plant downtime due to repair, an alternate manufacturing approach is required to fabricate the multi-metallic tubes with relative ease and better interface properties.

SUMMARY

Shear-assisted extrusion assemblies are provided. The assemblies can include: a billet assembly containing a billet comprising a billet outer material and a billet inner material in at least one cross-section; a tool operably engaged with the billet; an extrudate receiving channel configured to receive extrudate from the tool, wherein the extrudate comprises extruded outer material and extruded inner material in at least one cross-section, the extruded outer material being the same material as the billet outer material, and the extruded inner material being the same as the billet inner material.

Methods for producing a multi-material shear-assisted extrudate are also provided. The methods can include: providing a billet comprising billet inner material and billet

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outer material; providing shear-assisted force to a tool operably engaged with the billet to form an extrudate comprising extruded inner material and extruded outer material wherein the extruded outer material is the same material as the billet outer material, and the extruded inner material being the same as the billet inner material.

DRAWINGS

Embodiments of the disclosure are described below with reference to the following accompanying drawings.

FIG. 1 is an example depiction of a portion of a shear-assisted extrusion assembly according to an embodiment of the disclosure.

FIG. 2 is a cross-section of an extrudate or extruded material according to an embodiment of the disclosure.

FIG. 3 is an example configuration of a portion of a shear-assisted assembly according to an embodiment of the disclosure.

FIG. 4 depicts another configuration of a portion of a shear-assisted extrusion assembly according to an embodiment of the disclosure.

FIG. 5 is a depiction of a shear-assisted extrudate according to an embodiment of the disclosure.

FIG. 6 is an exploded view of a billet configuration for use with shear-assisted extrusion assemblies according to an embodiment of the disclosure.

FIG. 7 is an extrudate material according to an embodiment of the disclosure.

FIGS. 8-12 are example billet configurations and commensurate extrudate materials obtained utilizing the example billet configurations according to embodiments of the disclosure.

FIG. 13 is a depiction of data of extruded material according to an embodiment of the disclosure.

FIG. 14 is a depiction of the operating conditions of a shear-assisted extrusion assembly utilizing differing materials according to an embodiment of the disclosure.

FIG. 15 is another example of shear-assisted extrusion assembly parameters according to an embodiment of the disclosure.

FIG. 16 is an example of co-metallic extruded material properties according to an embodiment of the disclosure.

FIGS. 17A-17E are depictions of multi-metallic extruded material according to an embodiment of the disclosure.

FIGS. 18A-18C are depictions of interfacial portions of multi-metallic extruded materials according to an embodiment of the disclosure.

FIGS. 19A-19D are depictions of multi-metallic extruded materials according to an embodiment of the disclosure.

FIGS. 20A-20E are depictions of interfacial multi-metallic extruded materials according to an embodiment of the disclosure.

FIGS. 21A-21C are depictions of portions of multi-metallic extruded materials according to an embodiment of the disclosure.

FIGS. 22A-22D are depictions of multi-metallic extruded materials according to an embodiment of the disclosure.

FIGS. 23A-23B are depictions of a billet configuration according to an embodiment of the disclosure.

FIGS. 24A-24B are depictions of billet configurations according to an embodiment of the disclosure.

FIGS. 25A-25C are depictions of multi-metallic extruded materials according to an embodiment of the disclosure.

FIGS. 26A-26B are depictions of multi-metallic extruded materials according to an embodiment of the disclosure.

FIG. 27 is a depiction of a multi-metallic extrudate according to an embodiment of the disclosure.

FIG. 28 is a depiction of a multi-metallic extrudate according to an embodiment of the disclosure.

DESCRIPTION

This disclosure is submitted in furtherance of the constitutional purposes of the U.S. Patent Laws “to promote the progress of science and useful arts” (Article 1, Section 8).

The present disclosure will be described with reference to FIGS. 1-28. The present disclosure provides assemblies and methods for creating a co-extruded multi-metallic extrudates that can be co-extruded in the form of tubes. The extrudates can include at least an extruded inner material within an extruded outer material. The inner material can be referred to as a core and the outer material can be referred to as a cladding or sleeve. In tube configurations, the inner material can be referred to as an inner layer and the outer material as an outer layer.

The materials (inner, outer, intermediate) that make up the extrudate can have distinct chemical and/or mechanical properties. For example, co-extrusion of 6061 (shell) and 7075 (core), 1100 (shell) and 7075 (core), and 1100 (shell) and 2024 (core) was performed to complete extrudates having commensurate outer and inner materials having sound bonding at the interface. Furthermore, thickness of the sleeve and core can be controlled via the area ratio of the constituent billet material. Multi-metallic tubes can be fabricated utilizing a tool that engages a mandrel.

The ShAPE process works in billet area ratio of 1/3 and 2/3, and also 1/2 and 1/2 leading to the tube thickness of 1/3 and 2/3, and 1/2 and 1/2, respectively. Based on the current observations, a much thinner core/sleeve, preferably having 1/4 of the tube thickness is quite possible. Therefore, either a thicker or thinner coating can easily be fabricated. Temperature control of the process using the shear and friction parameters can provide for more flexible fabrication of multi-metallic components having significantly different flow stresses. Furthermore, multi-metallic extrudate tubes exhibiting variable thicknesses can be fabricated by controlling the billet stack with minimal effort in billet fabrication. This allows for joining the bi-metallic tubes to other structures so that easily weldable material of necessary thickness is the sleeve. Difficult to extrude material combination such as 1100 and 7075/2024 Al can easily be co-extruded.

Furthermore, these multi-metallic extrudates, having gone through high temperature severe plastic deformation, can be aged without solutionizing or annealing heat treatments, thereby thermal stresses and the subsequent dimensional instability can be avoided and/or grain growth minimized.

Referring to FIG. 1, a portion of an example shear-assisted extrusion apparatus 10 is shown that includes a billet holder assembly 12 that comprises and container base 12a and a container ring 12b. Within assembly 12 is billet 20. Billet 20 comprises at least two materials, but may contain additional materials in different alignments. In this configuration, billet material 20 comprises at least a billet inner material 22b and a billet outer material 24b. As an example, materials 22b and 24b will be different materials as noted herein. The materials may differ in chemical and/or mechanical properties. The materials may also have different dimensions. For example, the thickness ratio of the materials as shown appear to be the same. Other thickness ratios can be utilized with the outer material being less or greater thickness than the inner material.

Tool 14 can be retained within tool holder 21 and operably engaged with billet material 20 to create a high shear region 26 at die face 28. As shown, a rotational force and axial force is applied to die face 28 of tool 14. The axial force may be applied from the tool upon the feed material; alternatively, the axial force may be applied from the feed material upon the tool. In accordance with example implementations, the shear assisted extrusion process will provide an extrudate 18. Mandrel 16 can be anchored in billet holder assembly 12 and extend through billet material 20 and die face 28 to extrudate receiving channel 11.

This application is related to U.S. Continuation-In-Part patent application Ser. No. 17/473,178 filed Sep. 13, 2021, entitled “Devices and Methods for Performing Shear-Assisted Extrusion and Extrusion Processes”, the entirety of which is hereby incorporated by reference herein.

Referring to FIG. 2, extrudate 18 can comprise at least extruded inner material 22e and extruded outer material 24e. These extruded materials (22e/24e) can be the same as billet materials (22b/24b) and have substantially the same ratio of thickness (e.g., area ratio of 1/3 and 2/3, and also 1/2 and 1/2 leading to the tube thickness of 1/3 and 2/3, and 1/2 and 1/2, respectively). The extruded materials can be bonded at interface 23 while the billet materials are not bonded.

Referring to FIGS. 3-12, various combinations of billet material configurations as well as extruded material configurations are shown. For example, FIGS. 3-7 demonstrate inner billet material 22b and outer billet material 24b bounded by billet material 40b. When shear-assisted extruded, extruded inner material 22e and outer extruded outer material 24e are bounded by extruded material 40e. This extruded material can have bonding interfaces between the inner 22e and outer 24e materials as well as between the outer 24e and extruded material 40e and/or the inner 22e and extruded material 40e.

Referring to FIGS. 8-12 a summary of example billet and extruded material configurations are shown. This summary is not provided to be exhaustive, just as an example. Accordingly, the billet material and the extrudate can include intermediate materials 90b/90e. These intermediate materials can include multiple materials themselves. Thus, the materials can include multiple layers in at least one cross-section. Additionally, with reference to FIGS. 10 and 11, inner and/or outer materials can extend to complete and be the same material as bounding materials. Accordingly, the inner or outer materials can be the same as bounding materials (e.g., 40). Also, with reference to FIG. 12, multiple inner and multiple outer materials can be provide as shown. Accordingly, outer materials can include distinct materials 24 and 120. Additionally, inner materials can include distinct materials 22 and 122. As the depiction provides, the abutment of these materials need not align.

Accordingly, billet containing assembly 12 containing a billet that includes a billet outer material 22b and a billet inner material 24b in at least one cross-section. The billet 20 can also include a billet intermediate material 90b between billet inner material 22b and billet outer material 24b in the at least one cross-section. The assembly can include mandrel 16 extending from billet containing assembly 12 to tool 14. Mandrel 16 can extend through an opening in die face 28 of the tool 14. Billet material 20 can also include a billet boundary material 40b directly adjacent and/or lateral of either billet inner material 22b or outer material 24b. the billet boundary material is the same material as the billet outer material. Billet boundary material 40b can be a different material than either or both of the billet inner material 22b or outer material 24b. In at least one cross-section the

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thickness of one or more of the inner **22e**, outer **24e**, and/or intermediate **90b** billet materials is different. Each material thickness can be as low as a 10^{th} of the thickness of the material directly adjacent thereto. In billet container assembly **12**, the billet materials can be unbound.

The extrudate **18** can also include extruded intermediate material **90e** between extruded outer material **24e** and extruded inner material **22e** in the at least one cross-section.

In accordance with the above assemblies, the methods can include a billet intermediate material **90b** between the billet inner material **22b** and billet outer material **24b**. Billet intermediate material **90b** can be a single material or multiple different materials. Inner **22b** and outer **24b** billet material can be about the same thickness in at least one cross-section. Extruded inner **22e** and outer **24e** materials can be about the same thickness in at least one cross-section. Inner billet material **22b** can be 1/10 to 9/10 the thickness of outer **24b** billet material. Extruded inner material **22e** can be 1/10 to 9/10 of the thickness of extruded outer billet material **24e**. The extruded materials define bonded interfaces between different and previously unbound billet materials.

As an example method with reference to Table 1 below, multi-metallic extrudates were prepared. These parameters are not limiting and ranges of rotational speed as well as advance rate can be utilized.

TABLE 1

Summary of example shear-assisted processing conditions employed to fabricate the multi-material extrudates.			
Material combination	Die rotational speed (RPM)	Die advance rate (mm/min)	Tube length, mm
1100 (24) and 7050 (22) Al	40	60	1960
1100 (24) and 2024 (22) Al	45		1830

The extruded tubes were sectioned for various microscopy and mechanical property characterizations in as-extruded condition. For optical and scanning electron microscopy (SEM) analysis, the tubes were sectioned along both longitudinal and transverse cross-sections. The cut samples were mounted in an epoxy, polished to a surface finish of 0.05 μm using colloidal silica suspension, and etched using Keller's reagent. Both stereo and light microscopy (in bright field mode) analyses of the samples were carried to analyze the tube quality and interface characteristics. The samples were repolished to 0.05 μm surface finish for SEM and energy-dispersive X-ray spectroscopy (EDS) analysis. SEM in back scattered electron (BSE) mode and EDS analyses of the clad tubes and the interface between 1100/7075 Al, and 1100/2024 Al were completed using the FEI Quanta 3D field emission gun dual beam FIB/SEM. Vickers hardness measurements in as-extruded was carried out using Clark Microhardness tester CM-700AT and five indents per sample was done. Tensile testing of 127 mm (5") long tubes was carried out in accordance with ASTM-B557 standard. All the results and discussion are presented in the next section.

Referring to FIG. 3, and with reference to Table 2, tensile properties of multi-metallic extrudates having differing inner and outer thickness ratios are shown.

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TABLE 2

Material combinations	YS (MPa)	UTS (MPa)	EI %
5 6061/7075 Al ($\frac{1}{2}$ and $\frac{1}{2}$)	175	252	13
6061/7075 Al ($\frac{2}{3}$ and $\frac{1}{3}$)	168	224	8.5
1100/7075 Al	132	219	9
1100/2024 Al	116	198	11.5

The measured tool temperature ($^{\circ}\text{C}$.) and the spindle torque (Nm) as a function of plunge depth for both the extrusions is presented in FIGS. 13 and 14, respectively. What can be considered a long steady-state temperature region was followed by an initial transient temperature region.

Steady-state temperatures in 1100/7075 Al and 1100/2024 Al were about 380 and 390 $^{\circ}\text{C}$., respectively, mainly due to slightly lower rotational speed in the former extrusion. An initial transition in spindle torque was also noted (FIG. 15) followed by an almost steady-state. However, beyond plunge depth of ~40 mm, oscillation in torque was observed. Exact origin of these oscillations is not known; however, a possible hypothesis could be based on the sticking characteristics of 1100 Al exhibits during thermomechanical processing. During ShAPE process, 1100 Al sticks to the extrusion die and at higher force the material slips, then the continuation of this process would result in torque oscillations, as noted in FIG. 15.

Referring next to FIG. 16, a comparison of coextruded multi-metallics is shown. Additional analysis of coextruded materials have been performed using Stereo and light microscopy of clad 1100 on 7075 Al and 1100 on 2024 Al are presented in FIGS. 17A-17E and 20A-20E respectively. FIGS. 17A-17B and 20A-20B represent the macro-view of both longitudinal and transverse cross-sections for clad 1100/7075 Al and 1100/2024 Al, respectively. The clad outer layer (1100 Al) was continuously present along the tube (dark region) in all the conditions and cross-sections analyzed. The bright inner region in FIGS. 17A-17B and 20A-20B correspond to 7075 Al and 2024 Al, respectively. Thickness of tubes measured on the transverse cross-section for 1100/7075 and 1100/2024 are 0.92 ± 0.05 mm and 0.97 ± 0.14 mm, respectively. As noted, tube thickness in 1100/2024 Al was higher than 1 mm in certain locations as compared with 1100/7075 Al. The thickness of the outer and inner layers for 1100/7075 Al are 0.6 ± 0.1 and 0.36 ± 0.06 mm, respectively, and thickness for the outer and inner layers for 1100/2024 Al are 0.57 ± 0.1 and 0.38 ± 0.03 mm, respectively. In both the tubes, the 1100 Al outer layer exhibited larger thickness than the expected 0.5 mm based on the billet area. The light microscopy images are presented in FIGS. 17B-17E and 20B-20E. Both outer and inner layers of the tube for both clad configurations did not have any processing flaws. Furthermore, higher magnification analysis of the interface region shows that the interface in both the conditions is defect-free and wavy in nature.

SEM-BSE and SEM-EDS analyses of 1100/7075 and 1100/2024 Al clad tubes are presented in FIGS. 18A-18C and 21A-21C, respectively. Interface region between 1100/7075 and 1100/2024 Al clad tubes is marked with a solid arrow. For the most part, interface region did not exhibit discontinuities except for a few micro-cracks as marked by the dotted arrow in FIG. 18A. Therefore, the interface region was metallurgically bonded in both the conditions as clearly observed in FIGS. 18B and 21B.

Furthermore, presence of second phases is noted in both 7075 and 2024 Al alloys. Results of SEM-EDS line scan analysis of 1100/7075 Al and 1100/2024 Al are presented in FIGS. 18C and 21C, respectively. As expected, Zn and Cu content increased across the clad interface from 1100 Al into the 7075 and 2024 Al, respectively. Additional data are provided in Tables 3-4 below regarding hardness and tensile strength of the multi-metallic materials.

TABLE 3

Hardness summary in as-extruded condition for 1100 and 7075 Al and 1100 and 2024 Al clad tubes		
	Material identification	Hardness (HV0.1)
1100/7075 Al	1100 Al	20.2 ± 0.2
	7075 Al	83.3 ± 3.0
1100/2024 Al	2024 Al	68.0 ± 1.0
	1100 Al	21.0 ± 0.3

TABLE 4

Tensile properties Summary of the clad Al tubes.			
	YS (MPa)	UTS (MPa)	El. %
1100/7075 Al	135 ± 3.5	213 ± 5	6.4 ± 1.4
1100/2024 Al	117 ± 5.4	199 ± 4	10.6 ± 1.1

Overall, the presence of metallurgically bonded interface obtains clad tubes with good structural integrity.

Referring next to FIGS. 19A-19D and 22A-22D, inner and outer materials having similar cross-section area are provided and metallurgical bonding was determined to be present in both cases of 1100 outer and 7075 or 2024 inner.

In accordance with an example implementation, at least a pair of billet designs are shown in FIGS. 23A-23B and 24A-24B. FIGS. 23A-23B demonstrate a similar thickness while FIGS. 24A-24B demonstrate an inner 1/3 to outer 2/3 ratio. The materials used were inner 7075 and outer 6061. In FIGS. 25A-28 analysis results of both FIGS. 23A-B and 24A-B designs are shown.

Successful extrudate fabrication of distinctly different flow strength materials can provide heat exchangers with high temperature strength materials having a corrosion/oxidation resistance material as either an inner or outer material. At least one example of this application is co-extrusion of copper-nickel bi-metallic tubing which has been performed using these methods. Moreover, the apparatus and methods described herein enable continuous cladding from start to stop. For example, the outer material can be present over an entirety of an extrudate surface. Additionally, or alternatively, the apparatus and method provide the ability to vary or change outer material thickness. That is, a thicker and/or thinner outer material may be continuously fabricated along an outer surface of the extrudate.

In compliance with the statute, embodiments of the invention have been described in language more or less specific as to structural and methodical features. It is to be understood, however, that the entire invention is not limited to the specific features and/or embodiments shown and/or described, since the disclosed embodiments comprise forms of putting the invention into effect. The invention is, therefore, claimed in any of its forms or modifications within the proper scope of the appended claims appropriately interpreted in accordance with the doctrine of equivalents.

The invention claimed is:

1. A shear assisted extrusion assembly, the assembly comprising:

a billet containing assembly configured for receiving a billet comprising a billet outer material and a billet inner material in at least one cross-section;

a tool configured to operably engage the billet outer material and the billet inner material to cause plastic deformation of the billet outer material and the billet inner material; and

an extrudate receiving channel configured to receive extrudate from the tool, the extrudate comprising extruded outer material and extruded inner material in at least one cross-section;

wherein the extruded outer material comprises the same material as the billet outer material, and the extruded inner material comprises the same material as the billet inner material.

2. The assembly of claim 1 wherein the extruded inner material and the extruded outer material are bonded at an interface therebetween.

3. The assembly of claim 2 wherein:

further comprising the billet; and

wherein the billet further comprises a billet intermediate material between the billet outer material and the billet inner material in the at least one cross-section; and

the extrudate further comprises extruded intermediate material between the extruded outer material and the extruded inner material in the at least one cross-section.

4. The assembly of claim 1 further comprising a mandrel extending from the billet containing assembly to the tool, wherein the mandrel extends through an opening in a die face of the tool.

5. The assembly of claim 1 further comprising the billet; and

wherein the billet further comprises a billet boundary material adjacent to either the billet inner material or the billet outer material.

6. The assembly of claim 5 wherein:

the billet boundary material is axial to the billet inner material; and

the billet boundary material is axial to the billet outer material.

7. The assembly of claim 5 wherein the billet boundary material is a different material than the billet outer material.

8. The assembly of claim 5 wherein the billet boundary material comprises a different material than either or both of the billet inner or outer materials.

9. The assembly of claim 5 wherein in at least one cross-section a thickness of the inner and outer billet materials is different.

10. The assembly of claim 1, further comprising the billet.

11. A method for producing a multi-material shear assisted extrudate, the method comprising:

positioning a billet to be operably engaged with a tool to cause plastic deformation of the billet, the billet comprising a billet inner material and a billet outer material; and

establishing rotation and axial force between the tool and the billet to engage the tool with the billet to form an extrudate comprising extruded inner material and extruded outer material;

wherein the extruded outer material and the extruded inner material are bonded at an interface therebetween.

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12. The method of claim 11 wherein the extruded outer material comprises the same material as the billet outer material and the extruded inner material comprises the same as the billet inner material.

13. The method of claim 11 further comprising a billet intermediate material between the billet inner material and the billet outer material, wherein the billet intermediate material comprises a single material.

14. The method of claim 13 wherein the billet intermediate material comprises multiple different materials.

15. The method of claim 11 wherein the inner and outer billet material are about a same thickness in at least one cross-section.

16. The method of claim 15 wherein the extruded inner material and the extruded outer material are about the same thickness in at least one cross-section.

17. The method of claim 11 wherein the inner billet material is 1/10 to 9/10 the thickness of the outer billet material.

18. The method of claim 17 wherein the extruded inner material is 1/10 to 9/10 of the thickness of the extruded outer billet material.

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19. The method of claim 11 wherein the extruded inner and outer materials define bonded interfaces between different unbound billet materials.

20. A shear assisted extrusion assembly, the assembly comprising:

a billet containing assembly containing a billet, the billet comprising a billet outer material and a billet inner material in at least one cross-section;

a tool configured to operably engage the billet outer material and the billet inner material to cause plastic deformation of the billet outer material and the billet inner material;

a mandrel extending from the billet containing assembly to the tool, wherein the mandrel extends through an opening in a die face of the tool; and

an extrudate receiving channel configured to receive extrudate from the tool, the extrudate comprising extruded outer material and extruded inner material in at least one cross-section;

wherein the extruded inner material and the extruded outer material are bonded at an interface therebetween.

* * * * *